USE-CASE FLOW SPECIFICATION

Community Tool & Equipment Library

Use Case ID	UC-01
Use Case Name	Reserve / Checkout Tool
Primary Actor	Community Member
Supporting Actor	Library Custodian
Goal	Allow a member to reserve/check out available equipment while applying the required deposit and loan rules.

Preconditions

- The community member is registered and authorized to borrow equipment.
- The selected tool/equipment is available for the requested loan period.
- The member does not have an overdue item that prevents a new checkout under library policy.

Postconditions

- The selected tool is reserved/checked out for the approved loan duration.
- The required security deposit is calculated and recorded/held.
- The system records the transaction and makes the due/return date available to the member.

Main Success Scenario

1. The Community Member searches the tool and equipment inventory.
2. The system displays the selected item's availability and relevant tool/category details.
3. The member selects the required loan/return dates and requests a reservation or checkout.
4. The system validates the requested loan duration against the applicable limit.
5. The system calculates the required security deposit based on the tool category and loan duration.
6. The member confirms the checkout and deposit.
7. The system records the reservation/checkout and deposit hold.
8. The system updates the item's availability and displays a confirmation with the return date.
9. The use case ends successfully.

Alternate Flow – Member Has an Overdue Tool

A1.1 At Step 3, the system checks the member's current loans and finds an overdue tool.
A1.2 The system blocks the new checkout when the requested equipment is high-value or otherwise restricted by policy.
A1.3 The system informs the member that the checkout cannot proceed until the overdue item is returned/resolved.
A1.4 No new reservation is created and the inventory remains unchanged.

Business Rules

BR-01: Security deposits shall be calculated according to the tool/equipment category and applicable loan duration.
BR-02: Each tool category shall have a defined maximum loan duration.
BR-03: Late return fees shall be calculated when an item is returned after its due date.
BR-04: A member with overdue tools shall not be allowed to borrow additional high-value equipment.
BR-05: Damage assessments shall be recorded when returned equipment is found damaged.

Source basis: The scenario specifies checkout of power tools/camping gear/lawnmowers, deposit calculation, return dates, damage assessment, and rules for loan duration and late fees.